

LLM Evals

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{ 1 Online Metrics $\checkmark$ $\rightarrow$ domain/application specific
2 Offline Metrics $\rightarrow$ P/R/F1

3 Optional: What is your LLM latency

{ T F T T }

$\underbrace{\hspace{1.5cm}}_{\text{Time to First Token}}$

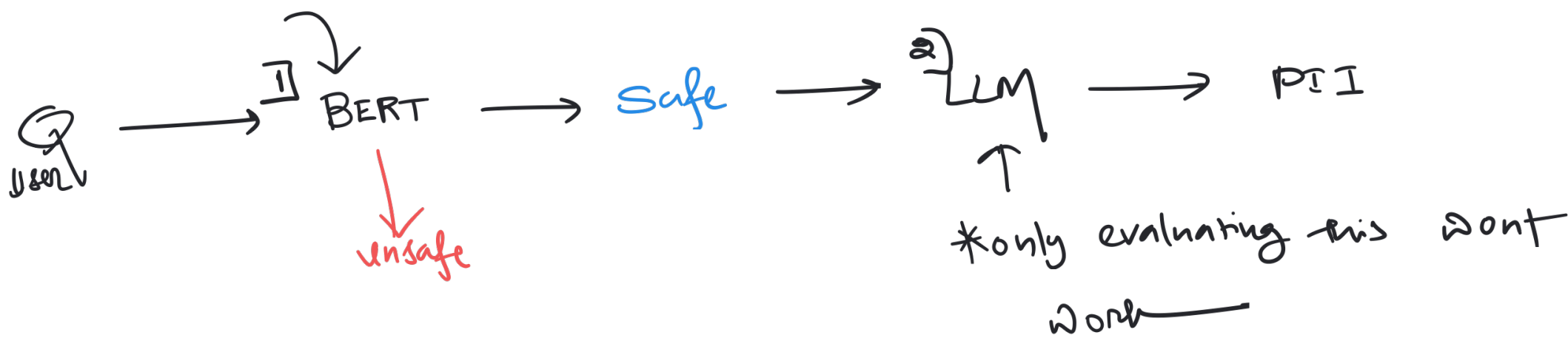
Inference optimization
vLLMs

$\checkmark$
Precision@k
Recall@k

"Dwell time"

LLM System eval

System:

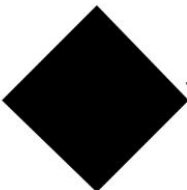


Simple LLM

Prompt

Summarize this:

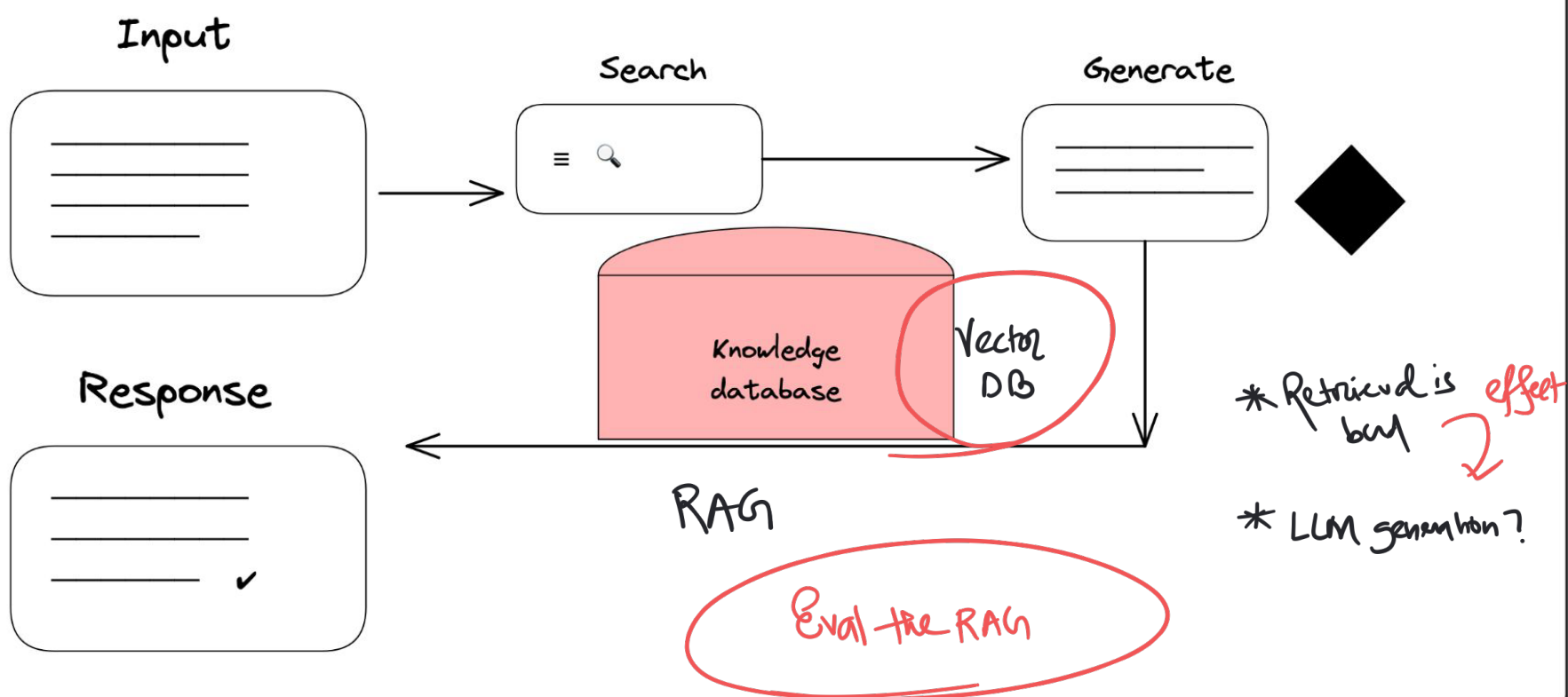
LLM



Output

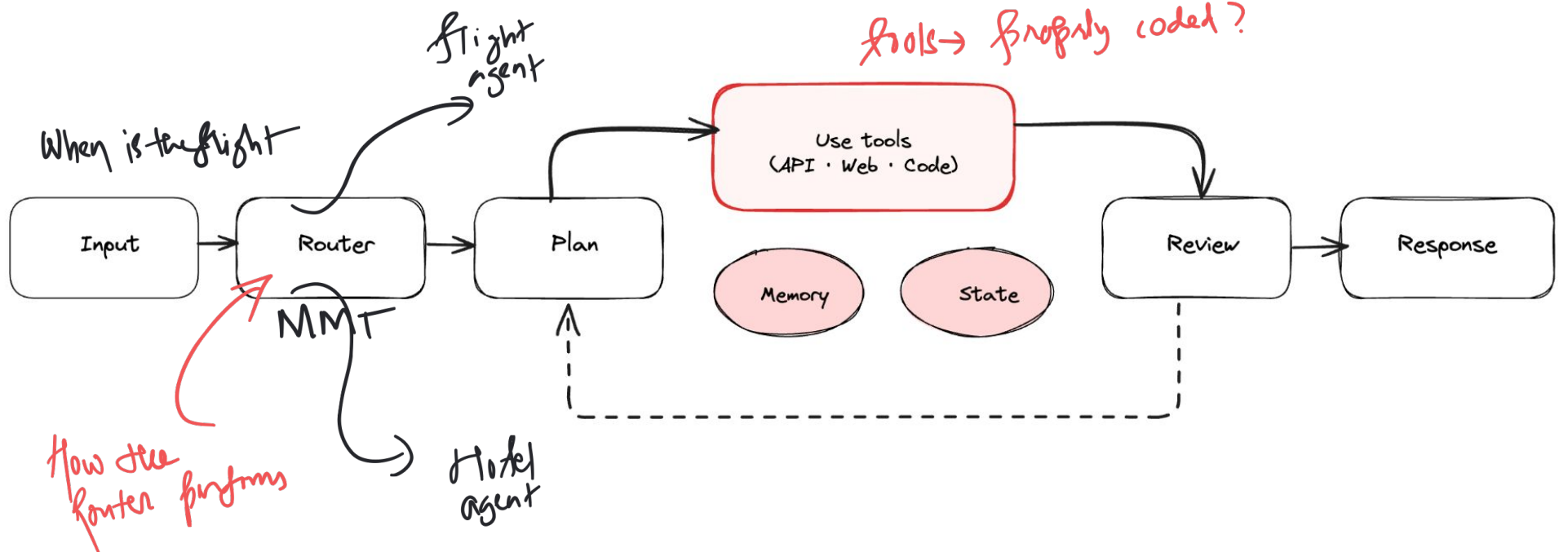
Here you go:

RAG



AI agents Eval the tools

tools → properly coded?



LLM App Eval — evaluate all the components
and not only the LLM.

Today

LLM Eval

LLM

Component

LLM-powered system

Python class
LLM Eval 2

class which we will have to the end of this module



VS.

✓ Guardrails

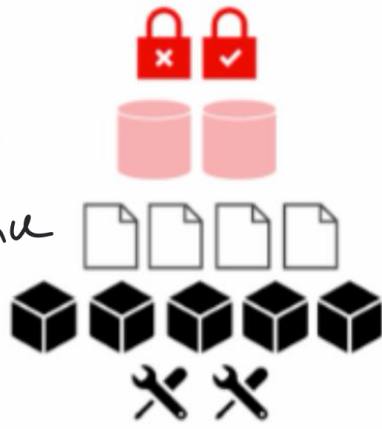
✓ Data sources

✓ Prompts

Models

✓ Tools

know



input

Expected y

predicted y

X

LL

Eval your LLM

→ Validation Data / Golden Dataset (this is based on your app)

1] Annotators (human intervention) 200-500 datapoints

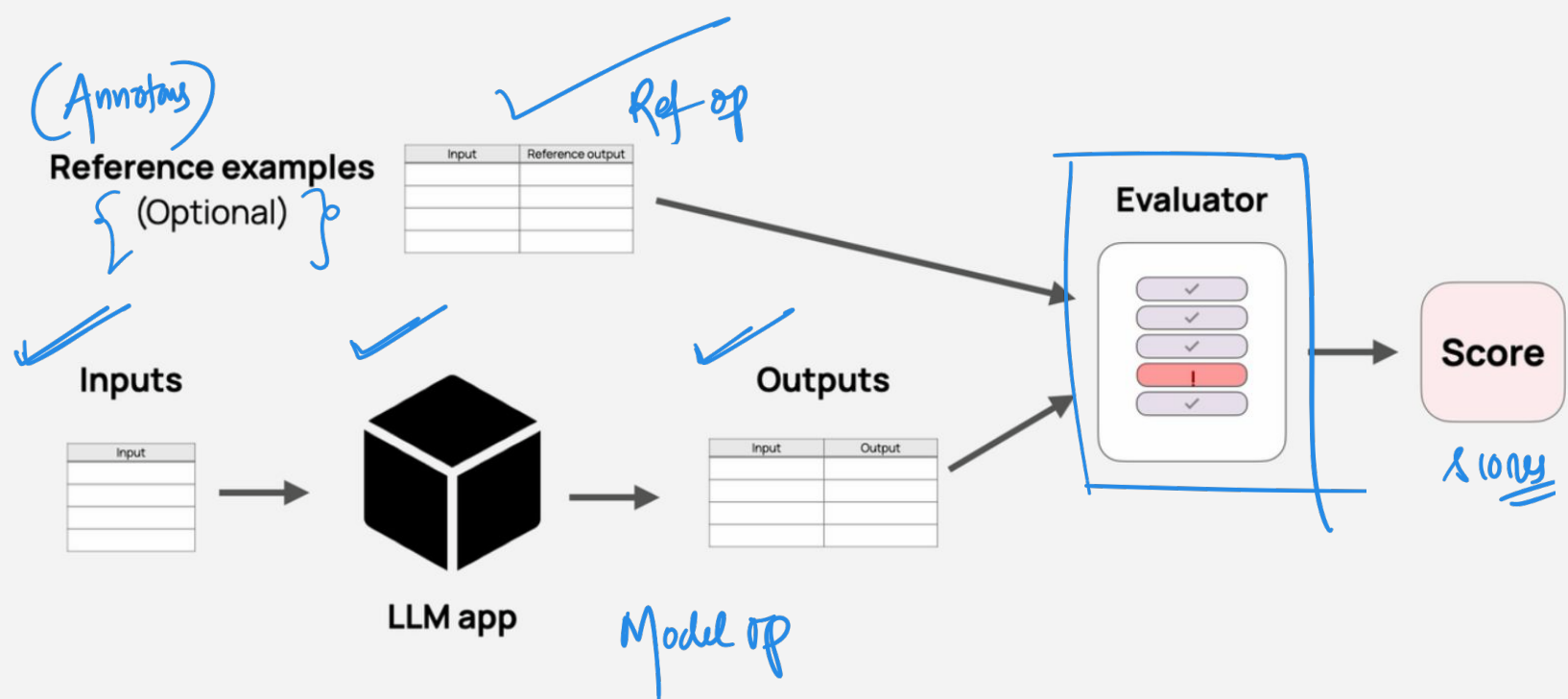
2] Real production data added to the original golden dataset.

✓✓
LLM → Machine translation (Eng - Hindi)
hire a Eng-Hindi translator

1 → English (sentence)
2 →

Hindi sentences

500 →



~~scribble~~

Tom is a cat

✓ Tom

is

a

feline ✓

~~scribble~~

1] Benchmarks $(100 \rightarrow 5/6)$

2] eval ✓ $(5/6 \rightarrow 1)$
app specific

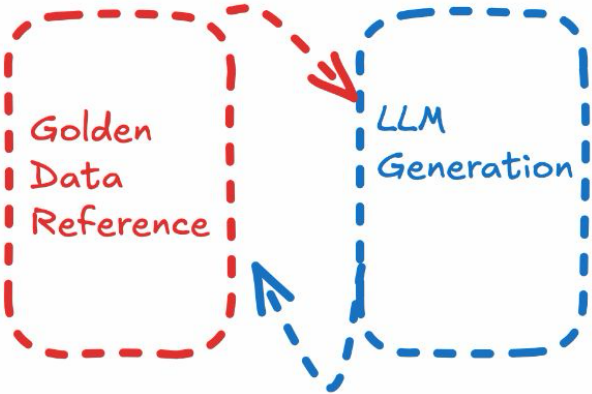
Do you have reference examples?

YES

NO

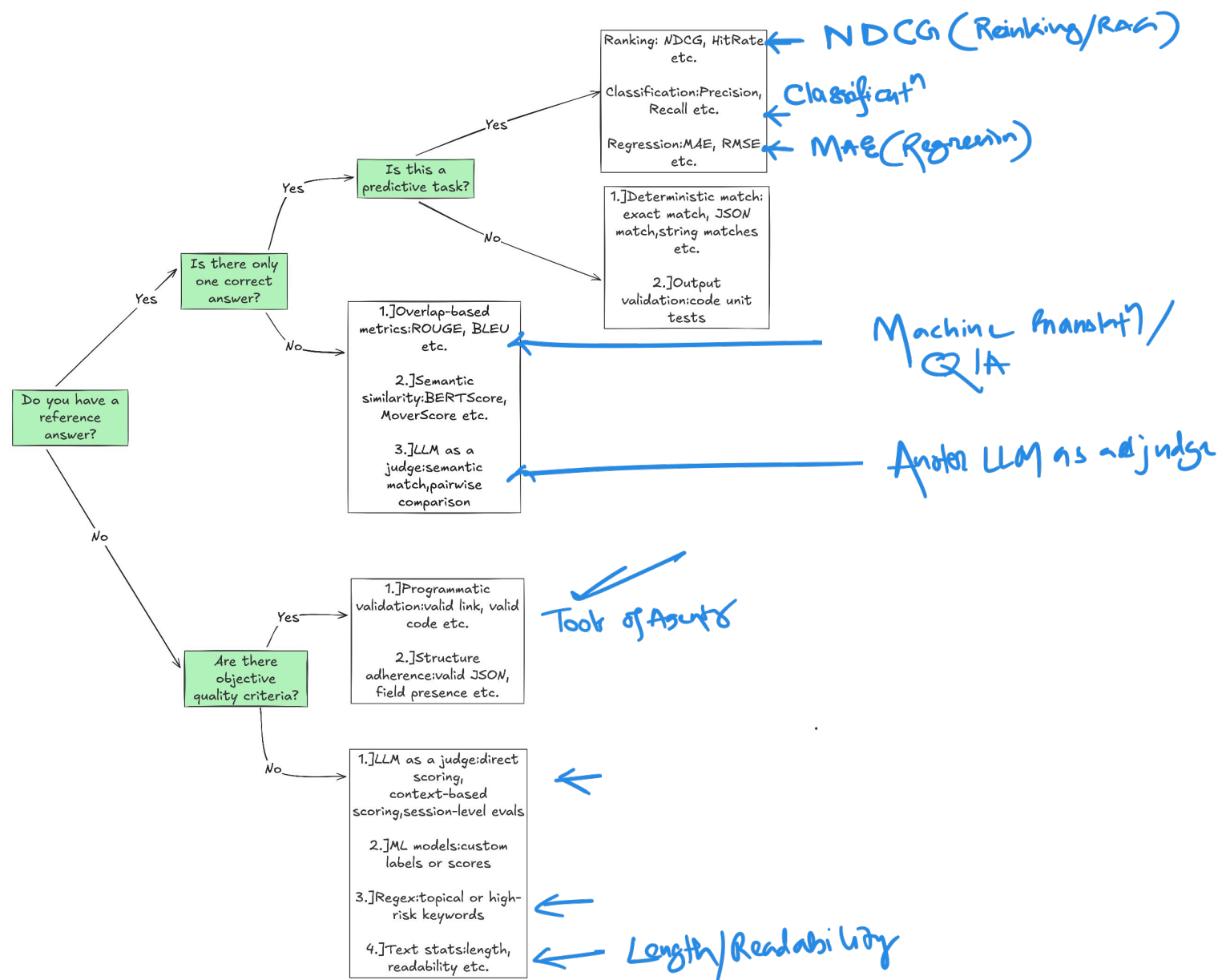
Reference-Based
Evaluation

Reference-Free
Evaluation



✓
input | Expected op

input | no frame
of
expected op



9:25

BLEU
ROUGE low scores → lack of semantic understanding

He often gets to school early



He often often gets
gets to to school
school early

Ref

He usually arrives early for class



He usually usually arrives
arrives early early for
for class

LLM
generated

Overlap = 0

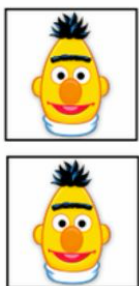
opik
by comet

BERTScore $\rightarrow$ Encoder Only Model (BERT etc)

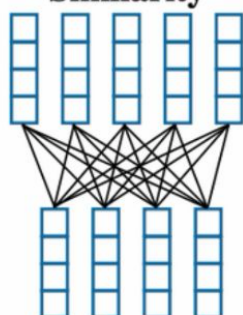
Ref
Reference $\mathcal{X}$
the weather is cold today

Candidate $\hat{\mathcal{X}}$
it is freezing today
LLM gen
op

Contextual
Embedding



Pairwise Cosine
Similarity



Maximum Similarity

Reference	it	is	freezing	today	idf weights
the	0.713	0.597	0.428	0.408	1.27
weather	0.462	0.393	0.515	0.326	7.94
is	0.635	0.858	0.441	0.441	1.82
cold	0.479	0.454	0.796	0.343	7.90
today	0.347	0.361	0.307	0.913	8.88
Candidate					

Importance Weighting
(Optional)

$$\rightarrow R_{\text{BERT}} = \frac{(0.713 \times 1.27) + (0.515 \times 7.94) + \dots}{1.27 + 7.94 + 1.82 + 7.90 + 8.88}$$

$\begin{bmatrix} 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 \end{bmatrix}$
 Cat sits $\rightarrow$ BERT $\rightarrow$

feline sitting $\rightarrow$ BERT $\rightarrow$
 $\begin{bmatrix} 0.9 & 0.1 \end{bmatrix}$ $\begin{bmatrix} 0.1 & 0.95 \end{bmatrix}$

Cat $\begin{bmatrix} \end{bmatrix}$

sits $\begin{bmatrix} \end{bmatrix}$

$\begin{bmatrix} \end{bmatrix}$
 feline

✓ Harmonic Mean
 BERTScore F1

✓ $\begin{bmatrix} \end{bmatrix}$

$\begin{bmatrix} \end{bmatrix} @ \begin{bmatrix} \end{bmatrix}$

$\uparrow$
 dot product

feline
 sitting

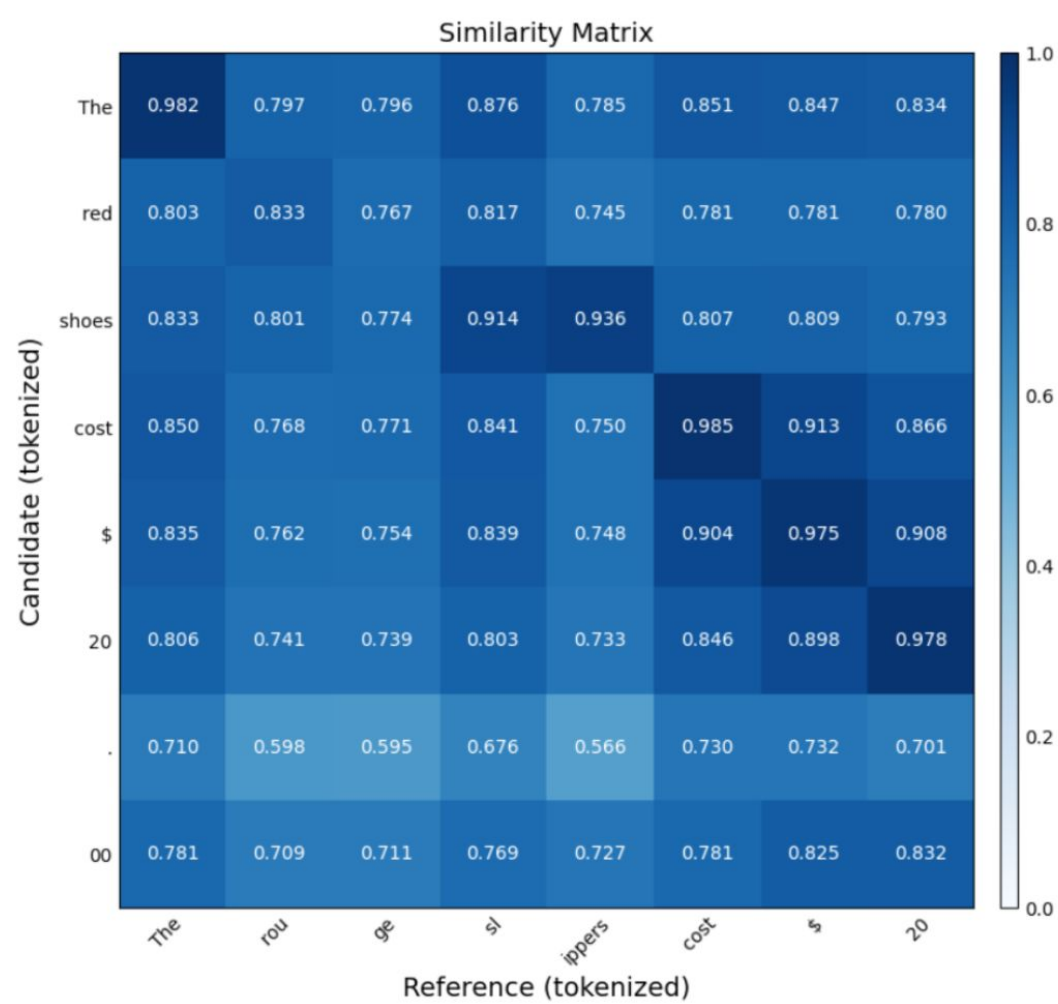
	Cat	sits $\leftarrow$ Reference
feline	0.99	0.11
sitting	0.10	0.99

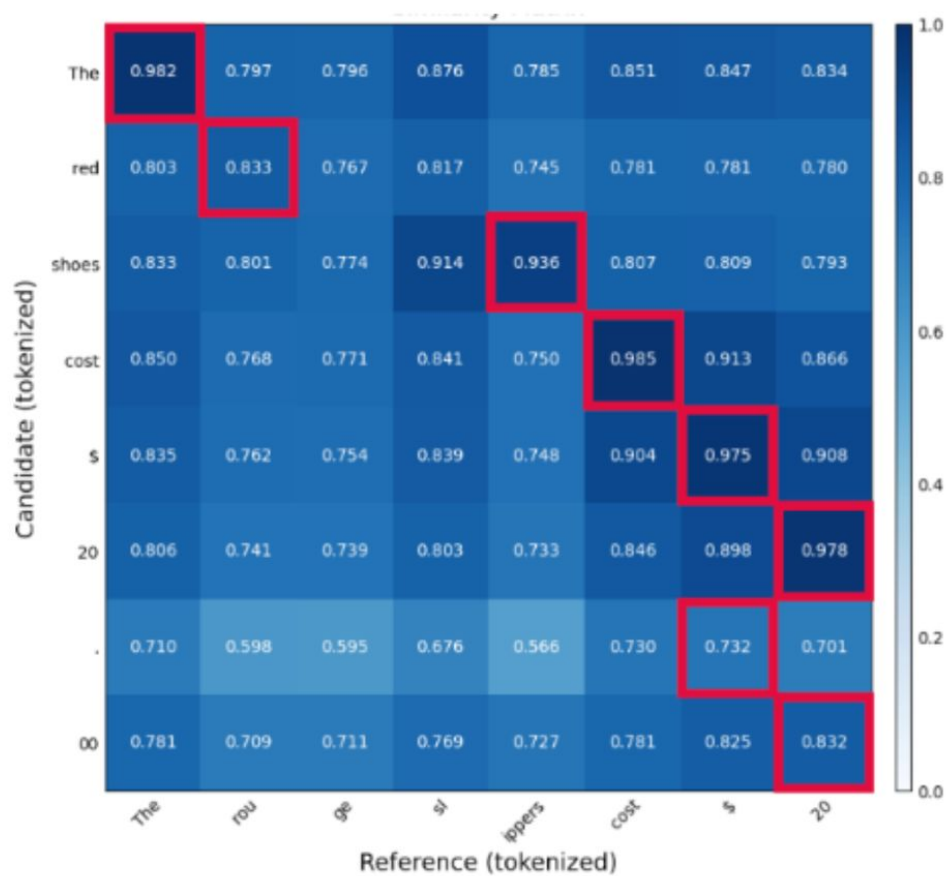
$\uparrow$
Candidate

$\begin{bmatrix} \end{bmatrix}$
 sitting

feline

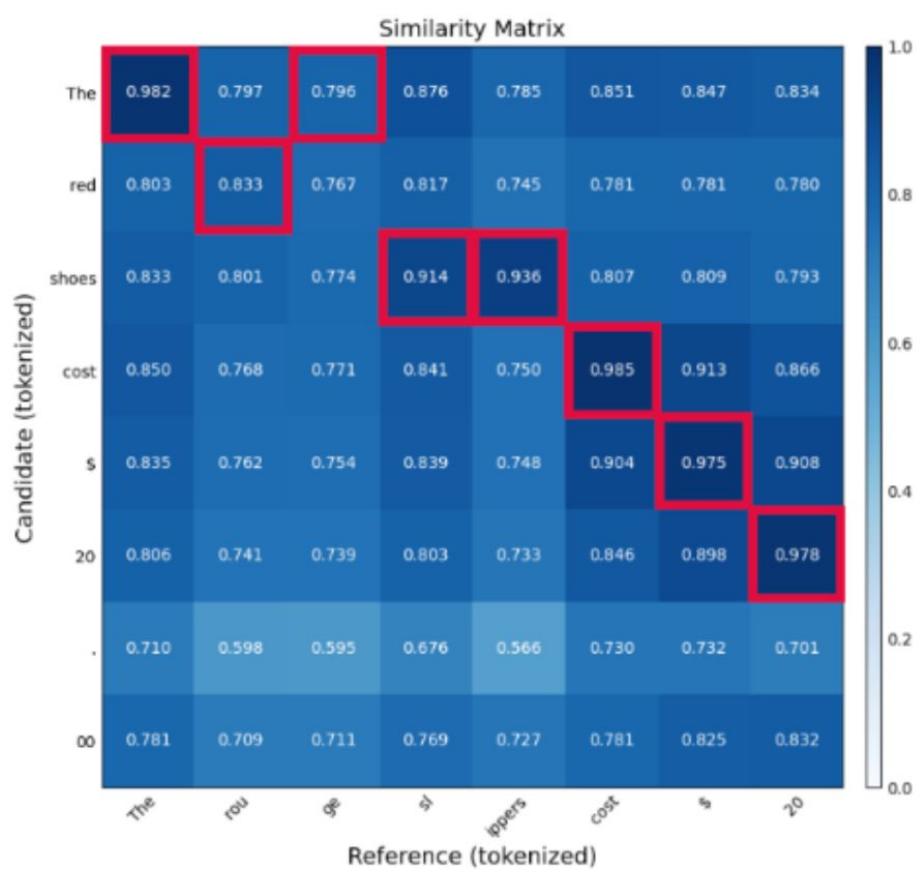
$$\left\{ \begin{array}{l} \text{BERTScore Precision} \frac{0.99 + 0.99}{2} \\ \text{BERTScore Recall} \frac{0.99 + 0.99}{2} \end{array} \right.$$





Maximum similarity scores of
candidate -> reference tokens:

**[0.982, 0.833, 0.936, 0.985,
 0.975, 0.978, 0.732, 0.832]**



Maximum similarity scores of
reference -> candidate tokens:

**[0.982, 0.833, 0.796, 0.914,
0.936, 0.985, 0.975, 0.978]**

BLEU

→ Precision

BLEU score stands for **B**ilingual **E**valuation **U**nderstudy.

When evaluating machine translation, multiple characteristics are taken into account:

- adequacy
- fidelity
- fluency

In its simplest form BLEU is the quotient of the matching words under the total count of words in **hypothesis** sentence (translation). Regarding the denominator BLEU is a precision oriented metric.

$$p_n = \frac{\sum_{n\text{-gram} \in \text{hypothesis}} \text{Count}_{\text{match}}(n\text{-gram})}{\sum_{n\text{-gram} \in \text{hypothesis}} \text{Count}(n\text{-gram})} = \frac{\sum_{n\text{-gram} \in \text{hypothesis}} \text{Count}_{\text{match}}(n\text{-gram})}{\ell_{\text{hyp}}^{n\text{-gram}}}$$

For example, the matches in the sample sentences are "to", "make", "people", "trustworthy", "to", "trust", "them"

reference → The way **to make people trustworthy** is **to trust them**
 $\ell_{\text{ref}}^{\text{unigram}} = 10$

hypothesis → **To make people trustworthy**, you need **to trust them**
 $\ell_{\text{hyp}}^{\text{unigram}} = 9$

$$p_1 = \frac{7}{9}$$

$$P_{\text{unigram}} = \frac{7}{9}$$

Unigram matches tend to measure adequacy while longer n-grams matches account for fluency.

Then precisions on various n-grams are aggregated using a weighted average of the logarithm of modified precisions.

$P_2 = \frac{\text{No. of matching bigrams}}{\text{total bigrams in LLM generated}}$

P_2

no. of words / n-grams in the LLM sen of

The, way, to, make, ...

To, make, people, trustworthy, ...

$P_2 =$

The way, way to, to make, make people, ...

Reference are from Golden Dataset

LLM generated output

"To make people"

No BP
BLEU score will be biased to small/short generation

The brevity penalty BP is function of the lengths of reference and hypothesis sentences.

$$BP = \begin{cases} 1 & \text{if } \ell_{\text{hyp}} > \ell_{\text{ref}} \\ e^{1 - \frac{\ell_{\text{ref}}}{\ell_{\text{hyp}}}} & \text{if } \ell_{\text{hyp}} \leq \ell_{\text{ref}} \end{cases}$$

LLM generated $>$ Reference length

LLM generated $\leq$ Reference length

$(e)^{1 - \frac{\text{Length of ref}}{\text{Length of LLM gen}}}$

BLEU example

Type	Sentence	Length
Reference	The way to make people trustworthy is to trust them.	$\ell_{\text{ref}}^{\text{unigram}} = 10$
Hypothesis	To make people trustworthy, you need to trust them.	$\ell_{\text{hyp}}^{\text{unigram}} = 9$

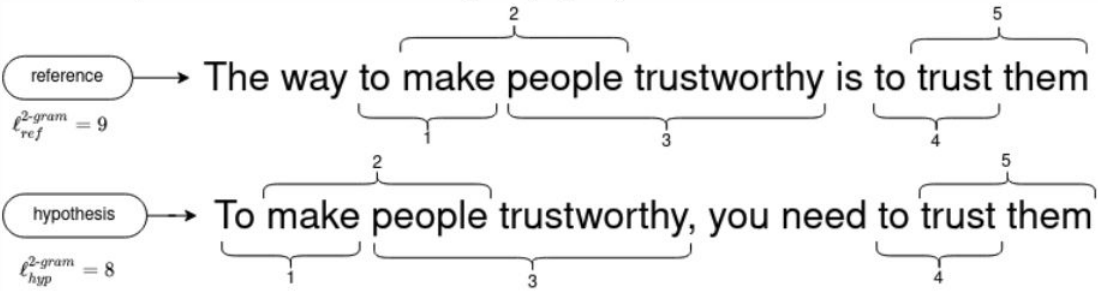
For this example we take parameters as the base line score, described in the paper, with $N = 4$, and a uniform distribution, therefore taking $w_n = \frac{1}{4}$.

$$\text{BLEU}_{N=4} = \text{BP} \cdot \exp \left(\sum_{n=1}^{N=4} \frac{1}{4} \log p_n \right)$$

$p_1 \quad p_2 \quad p_3 \quad p_4$

We then calculate the precision p_n for the different n-grams.

For instance, here is an illustration of the bigram (2-gram) matches



The following table details the precisions for 4 n-grams.

n-gram	1-gram	2-gram	3-gram	4-gram
p_n	$p_1 \frac{7}{9}$	$p_2 \frac{5}{8}$	$p_3 \frac{3}{7}$	$p_4 \frac{1}{6}$

$$\exp \left(\frac{1}{4} \cdot p_1 + \frac{1}{4} \cdot p_2 + \frac{1}{4} \cdot p_3 + \frac{1}{4} \cdot p_4 \right)$$

BLEU score ↓

no. of matched tokens

total no. of tokens in LLM gen

no. of matched tokens < total no. of tokens in LLM gen

HALUCINATION

ROUGE — Recall —

ROUGE score stands for **R**ecall-**O**riented **U**nderstudy for **G**isting **E**valuation.

Evaluation of summarization involves measures of

- coherence
- conciseness
- grammaticality
- readability
- content

In its simplest form ROUGE score is the quotient of the matching words under the total count of words in **reference** sentence (summurization). Regarding the denominator ROUGE is a recall oriented metric.

$$\text{ROUGE}_1 = \frac{\sum_{\text{unigram} \in \text{reference}} \text{Count}_{\text{match}}(\text{unigram})}{\sum_{\text{unigram} \in \text{reference}} \text{Count}(\text{unigram})} = \frac{\sum_{\text{unigram} \in \text{reference}} \text{Count}_{\text{match}}(\text{unigram})}{\ell_{\text{ref}}^{\text{unigram}}}$$

no. of matching tokens
total no. of tokens in reference

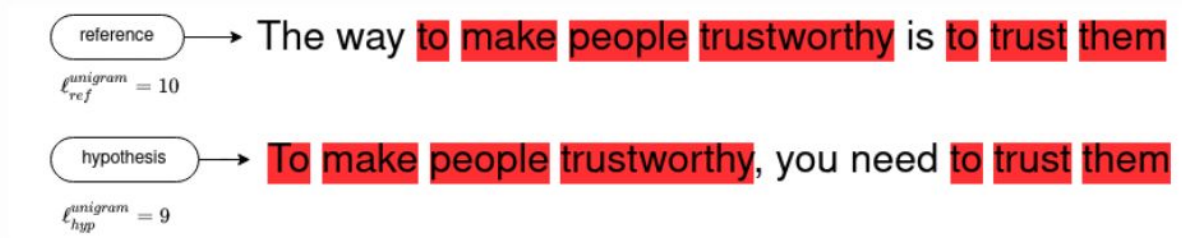
ROUGE-1 example

ROUGE-1 is the ROUGE-N metric applied with unigrams.

Type	Sentence	Length
Reference	The way to make people trustworthy is to trust them.	$\ell_{\text{ref}}^{\text{unigram}} = 10$
Hypothesis	To make people trustworthy, you need to trust them.	$\ell_{\text{hyp}}^{\text{unigram}} = 9$

The following illustrates the computation of ROUGE-1 on the summurization sentences

The following illustrates the computation of ROUGE-1 on the summarization sentences



$$ROUGE_1 = \frac{7}{10} = 0.7$$

Four ROUGE metrics are defined in the ROUGE paper: ROUGE-N, ROUGE-L, ROUGE-W and ROUGE-S. The next section presents the ROUGE-L score.

ROUGE-L

ROUGE-L or $ROUGE_{LCS}$ is based on the length of the longest common subsequence (LCS). To counter the disadvantages of a pure recall metric as in ROUGE-N, Rouge-L calculates the weighted harmonic mean (or f-measure) combining the precision score and the recall score.

The advantages of $ROUGE_{LCS}$ is that it does not require consecutive matches but in-sequence matches that reflect sentence level word order as n-grams. The other advantage is that it automatically includes longest in-sequence common n-grams, therefore no predefined n-gram length is necessary.

$$\begin{cases} R_{LCS} &= \frac{LCS(reference, hypothesis)}{\ell_{ref}^{unigram}} \\ P_{LCS} &= \frac{LCS(reference, hypothesis)}{\ell_{hypothesis}^{unigram}} \\ ROUGE_{LCS} &= \frac{(1+\beta^2)R_{LCS}P_{LCS}}{R_{LCS}+\beta^2P_{LCS}} \end{cases}$$

BLEU / ROUGE



BERT Score



dLNs as a judge

200

Q/A



Q

Ref A

GPT-mini Answer

X
prod

gpt-4o-large

⊕

JUDGE

eval

Judge

→ "Eval Prompt"

Flexibility

LLM as a judge (predefined)

Q A

satisfaction - 1 to 10

1 to 5

format - 1 to 5

Eval prompts



next Eval prompt

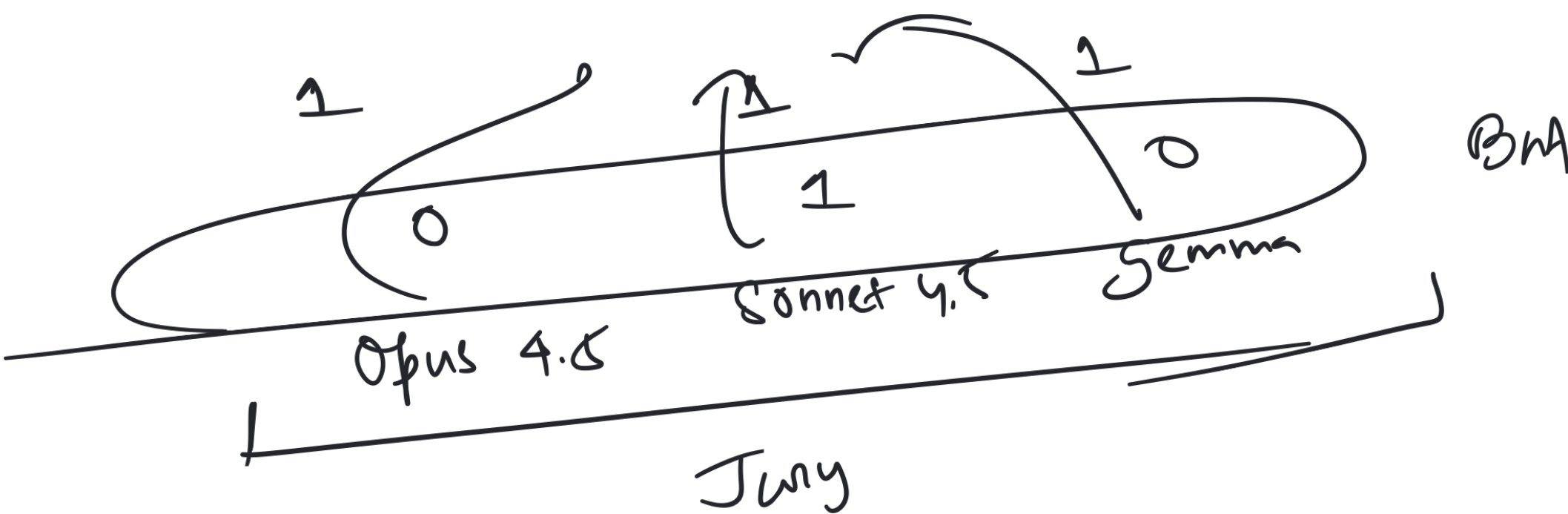
Jury of LLMs

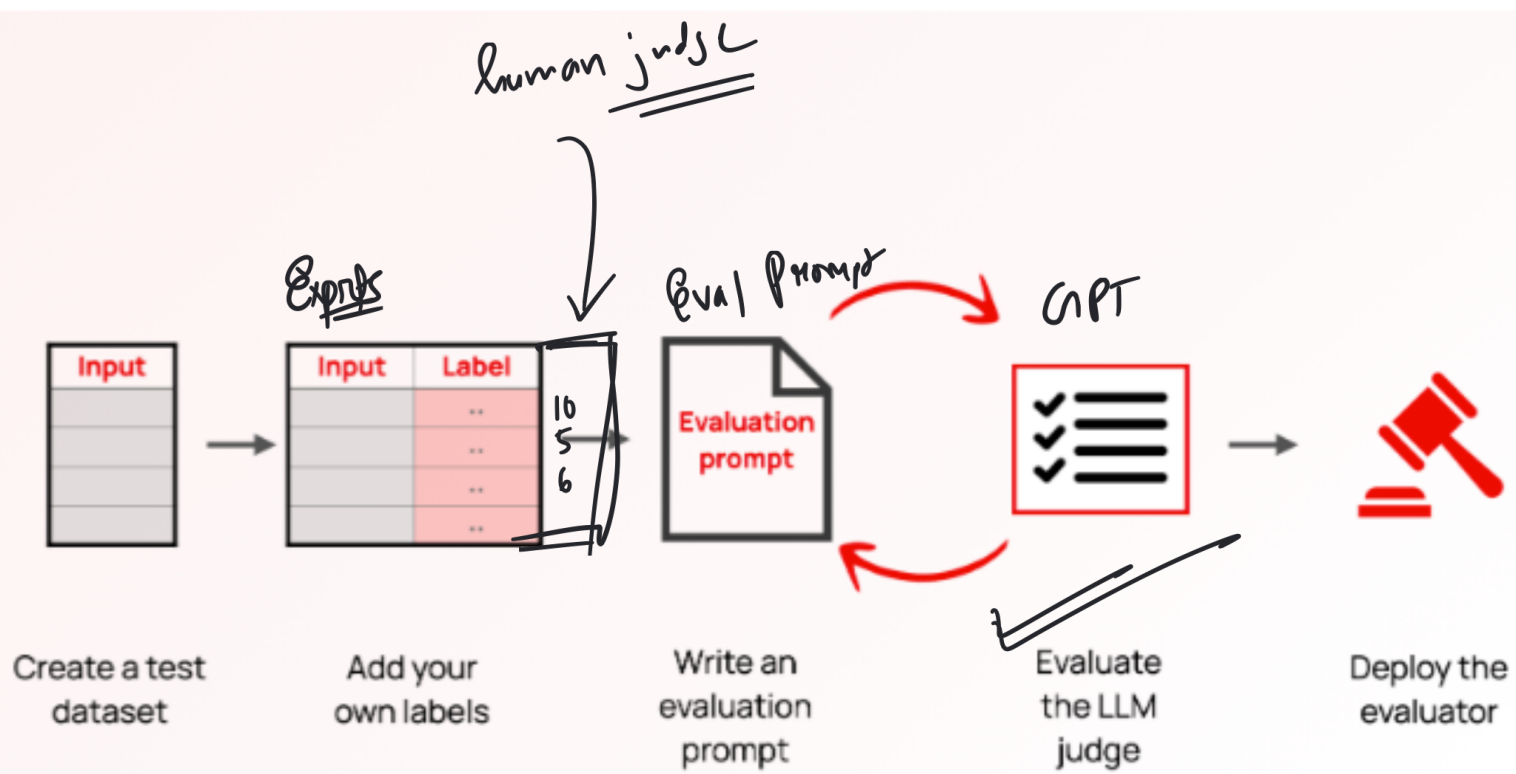
GPT 3.5 → generation

↑ judge

GPT 5

gpt5





R4

